

23. Construct a C program to implement first fit algorithm of memory management.

Code:

```
main.c
1  #include <stdio.h>
2
3  int main() {
4      int block[] = {100, 500, 200, 300};
5      int process = 150;
6      int i, flag = 0;
7
8      printf("Memory Blocks: 100 500 200 300\n");
9      printf("Process Size: %d\n", process);
10
11     for(i = 0; i < 4; i++) {
12         if(block[i] >= process) {
13             printf("Process Allocated to Block %d\n", i + 1);
14             flag = 1;
15             break;
16         }
17     }
18
19     if(flag == 0)
20         printf("Memory Not Allocated\n");
21
22     return 0;
23 }
```

Output:

```
Output
Memory Blocks: 100 500 200 300
Process Size: 150
Process Allocated to Block 2

=== Code Execution Successful ===
```